Problem A. Counting Roads

Time Limit 2000 ms
Mem Limit 262144 kB

Problem Statement

There are N cities and M roads. The i-th road $(1 \leq i \leq M)$ connects two cities a_i and b_i $(1 \leq a_i, b_i \leq N)$ bidirectionally. There may be more than one road that connects the same pair of two cities. For each city, how many roads are connected to the city?

Constraints

- $2 \leq N, M \leq 50$
- $1 \leq a_i, b_i \leq N$
- $a_i \neq b_i$
- All input values are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \ \ M \\
a_1 \ \ b_1 \\
: \\
a_M \ \ b_M
\end{array}
$$

Output

Print the answer in N lines. In the i-th line $(1 \leq i \leq N)$, print the number of roads connected to city i.

Sample 1

Input	Output
4 3 1 2 2 3 1 4	2 2 1 1

- City 1 is connected to the 1-st and 3-rd roads.
- City 2 is connected to the 1-st and 2-nd roads.
- City 3 is connected to the 2-nd road.
- City 4 is connected to the 3-rd road.

Sample 2

Input	Output
2 5 1 2 2 1 1 2 2 1 1 2	5 5

Sample 3

Input	Output
8 8 1 2 3 4 1 5 2 8 3 7 5 2 4 1 6 8	3 3 2 2 2 1 1 2

Problem B. Some Sums

Time Limit 2000 ms
Mem Limit 262144 kB

Problem Statement

Find the sum of the integers between 1 and N (inclusive), whose sum of digits written in base 10 is between A and B (inclusive).

Constraints

- $1 \leq N \leq 10^4$
- $1 \leq A \leq B \leq 36$
- All input values are integers.

Input

Input is given from Standard Input in the following format:

N A B

Output

Print the sum of the integers between 1 and N (inclusive), whose sum of digits written in base 10 is between A and B (inclusive).

Sample 1

Input	Output
20 2 5	84

Among the integers not greater than 20, the ones whose sums of digits are between 2 and 5, are: 2, 3, 4, 5, 11, 12, 13, 14 and 20. We should print the sum of these, 84.

Sample 2

Input	Output
10 1 2	13

Sample 3

Input	Output
100 4 16	4554

Problem C. Coloring Colorfully

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

N tiles are arranged in a row from left to right. The initial color of each tile is represented by a string S of length N.

The i-th tile from the left is painted black if the i-th character of S is `0`, and painted white if that character is `1`.

You want to repaint some of the tiles black or white, so that any two adjacent tiles have different colors.

At least how many tiles need to be repainted to satisfy the condition?

Constraints

- $1 \leq |S| \leq 10^5$
- S_i is `0` or `1`.

Input

Input is given from Standard Input in the following format:

S

Output

Print the minimum number of tiles that need to be repainted to satisfy the condition.

Sample 1

Input	Output
000	1

The condition can be satisfied by repainting the middle tile white.

Sample 2

Input	Output
10010010	3

Sample 3

Input	Output
0	0

Problem D. Low Elements

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Given is a permutation $P_1, \ldots, P_N$ of $1, \ldots, N$. Find the number of integers i $(1 \leq i \leq N)$ that satisfy the following condition:

- For any integer j $(1 \leq j \leq i)$, $P_i \leq P_j$.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $P_1, \ldots, P_N$ is a permutation of $1, \ldots, N$.
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

N
P_1 ... P_N

Output

Print the number of integers i that satisfy the condition.

Sample 1

Input	Output
5 4 2 5 1 3	3

$i = 1, 2$, and 4 satisfy the condition, but $i = 3$ does not - for example, $P_i > P_j$ holds for $j = 1$.

Similarly, $i = 5$ does not satisfy the condition, either. Thus, there are three integers that satisfy the condition.

Sample 2

Input	Output
4 4 3 2 1	4

All integers i $(1 \leq i \leq N)$ satisfy the condition.

Sample 3

Input	Output
6 1 2 3 4 5 6	1

Only $i = 1$ satisfies the condition.

Sample 4

Input	Output
8 5 7 4 2 6 8 1 3	4

Sample 5

Input	Output
1 1	1

Problem E. Comparison

Time Limit 2000 ms

Problem Statement

You are given two positive integers A and B. Compare the magnitudes of these numbers.

Constraints

- $1 \leq A, B \leq 10^{100}$
- Neither A nor B begins with a `0`.

Input

Input is given from Standard Input in the following format:

A
B

Output

Print `GREATER` if $A > B$, `LESS` if $A < B$ and `EQUAL` if $A = B$.

Sample 1

Input	Output
36 24	GREATER

Since $36 > 24$, print `GREATER`.

Sample 2

Input	Output
850 3777	LESS

Sample 3

Input	Output
9720246 22516266	LESS

Sample 4

Input	Output
12345678901234567890123456789 0 23456789012345678901234567890 1	LESS

Problem F. Palindromic Numbers

Time Limit 2000 ms
Mem Limit 262144 kB

Problem Statement

Find the number of *palindromic numbers* among the integers between A and B (inclusive). Here, a palindromic number is a positive integer whose string representation in base 10 (without leading zeros) reads the same forward and backward.

Constraints

- $10000 \leq A \leq B \leq 99999$
- All input values are integers.

Input

Input is given from Standard Input in the following format:

```
A B
```

Output

Print the number of *palindromic numbers* among the integers between A and B (inclusive).

Sample 1

Input	Output
11009 11332	4

There are four integers that satisfy the conditions: 11011, 11111, 11211 and 11311.

Sample 2

Input	Output
31415 92653	612

Problem G. Static Range Sum Queries

Time Limit 1000 ms
Mem Limit 524288 kB

Given an array of n integers, your task is to process q queries of the form: what is the sum of values in range $[a, b]$?

Input

The first input line has two integers n and q: the number of values and queries.

The second line has n integers $x_1, x_2, \ldots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has two integers a and b: what is the sum of values in range $[a, b]$?

Output

Print the result of each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
8 4 3 2 4 5 1 1 5 3 2 4 5 6 1 8 3 3	11 2 24 4

Problem H. Exception Handling

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

You are given a sequence of length N: $A_1, A_2, \ldots, A_N$. For each integer i between 1 and N (inclusive), answer the following question:

- Find the maximum value among the $N-1$ elements other than A_i in the sequence.

Constraints

- $2 \leq N \leq 200000$
- $1 \leq A_i \leq 200000$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
A_1 \\
\vdots \\
A_N
\end{array}
$$

Output

Print N lines. The i-th line ($1 \leq i \leq N$) should contain the maximum value among the $N-1$ elements other than A_i in the sequence.

Sample 1

Input	Output
3 1 4 3	4 3 4

- The maximum value among the two elements other than A_1, that is, $A_2 = 4$ and $A_3 = 3$, is 4.
- The maximum value among the two elements other than A_2, that is, $A_1 = 1$ and $A_3 = 3$, is 3.
- The maximum value among the two elements other than A_3, that is, $A_1 = 1$ and $A_2 = 4$, is 4.

Sample 2

Input	Output
2 5 5	5 5

Problem I. Lower

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

There are N squares arranged in a row from left to right.

The height of the i-th square from the left is H_i.

You will land on a square of your choice, then repeat moving to the adjacent square **on the right** as long as the height of the next square is not greater than that of the current square.

Find the maximum number of times you can move.

Constraints

- All values in input are integers.
- $1 \leq N \leq 10^5$
- $1 \leq H_i \leq 10^9$

Input

Input is given from Standard Input in the following format:

N
$H_1 \ H_2 \ \ldots \ H_N$

Output

Print the maximum number of times you can move.

Sample 1

Input	Output
5 10 4 8 7 3	2

By landing on the third square from the left, you can move to the right twice.

Sample 2

Input	Output
7 4 4 5 6 6 5 5	3

By landing on the fourth square from the left, you can move to the right three times.

Sample 3

Input	Output
4 1 2 3 4	0

Problem J. Candies

Time Limit 2000 ms
Mem Limit 262144 kB

Problem Statement

We have a $2 \times N$ grid. We will denote the square at the i-th row and j-th column ($1 \leq i \leq 2, 1 \leq j \leq N$) as (i, j).

You are initially in the top-left square, $(1, 1)$. You will travel to the bottom-right square, $(2, N)$, by repeatedly moving right or down.

The square (i, j) contains $A_{i,j}$ candies. You will collect all the candies you visit during the travel. The top-left and bottom-right squares also contain candies, and you will also collect them.

At most how many candies can you collect when you choose the best way to travel?

Constraints

- $1 \leq N \leq 100$
- $1 \leq A_{i,j} \leq 100$ ($1 \leq i \leq 2, 1 \leq j \leq N$)

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
A_{1,1} \ A_{1,2} \ \ldots \ A_{1,N} \\
A_{2,1} \ A_{2,2} \ \ldots \ A_{2,N}
\end{array}
$$

Output

Print the maximum number of candies that can be collected.

Sample 1

Input	Output
5 3 2 2 4 1 1 2 2 2 1	14

The number of collected candies will be maximized when you:

- move right three times, then move down once, then move right once.

Sample 2

Input	Output
4 1 1 1 1 1 1 1 1	5

You will always collect the same number of candies, regardless of how you travel.

Sample 3

Input	Output
7 3 3 4 5 4 5 3 5 3 4 4 2 3 2	29

Sample 4

Input	Output
1 2 3	5